



Flexible rechargeable battery (Credit: <https://ucsdnews.ucsd.edu>)

Sensor integrated in plants monitors arsenic levels in soil

Scientists from the Singapore-MIT Alliance for Research and Technology (SMART) have engineered a novel plant nano-bionic optical sensor that can detect and monitor, in real time, levels of the highly toxic heavy metal arsenic in the underground environment. This development provides significant advantages over conventional methods used to measure arsenic in the environment. It will be important for both environmental monitoring and agricultural applications to safeguard food safety, as arsenic is a contaminant in many common agricultural products such as rice, vegetables, and tea leaves.

The novel optical nano-sensors exhibit changes in their fluorescence intensity upon detecting arsenic. Embedded in plant tissues, with no detrimental effects on the plant, these sensors provide a non-destructive way to monitor the internal dynamics of arsenic taken up by plants from the soil. This integration of optical nano-sensors within living plants enables the conversion of plants into self-powered detectors of arsenic from their natural environment, marking a significant upgrade from the time- and equipment-intensive arsenic sampling methods of current conventional methods.



MIT arsenic sensor (Credit: <https://news.mit.edu>)

Stretchable micro-supercapacitors to self-power wearable devices



A team of international researchers, led by Huanyu "Larry" Cheng, Dorothy Quiggle Career Development Professor in Penn State's Department of Engineering Science and Mechanics, has developed a self-powered, stretchable system that will be used in wearable health-monitoring and diagnostic devices (Credit: Penn State College of Engineering, <https://news.psu.edu>)

Current versions of batteries and supercapacitors powering wearable and stretchable health-monitoring and diagnostic devices have many shortcomings, including low energy density and limited stretchability. But according to a research team, with members from Penn State, Minjiang University, and Nanjing University, a stretchable system that can harvest energy from human breathing and motion can be used in wearable health-monitoring devices.

Micro-supercapacitors are energy storage devices that have a small footprint, high power density, and the ability to charge and discharge quickly. However, when fabricated for wearable devices, conventional micro-supercapacitors have a sandwich-like stacked geometry that displays poor flexibility, long ion diffusion distances, and a complex integration process. So, an arrangement of micro-supercapacitor cells in a serpentine, island-bridge layout allows the configuration to stretch and bend at the bridges, while reducing deformation of the micro-supercapacitors, providing increased stretchability and adjustable voltage outputs needed to power a wearable device.